Mingjing Zhou

Email: zhoumingjingcqu@163.com | Mobile: +86-19112163571 | Page: mingmingjing30.github.io
Address: 5-4 Building 1, No. 28 Tielu Yicun, Jiulongpo District, Chongqing City, China

EDUCATION

Chongqing University Chongqing, China
B.Eng. (expected) in Electrical Engineering and Automation, CQU-UC Joint Co-op Institute *09/2022 – 06/2027*

- GPA: **88.12**/100
- Award: Team Captain, Successful Participant ("S" Award) at the U.S. Mathematical Contest in Modeling (MCM) (03/2024 & 03/2025)
  Third Prize of Excellent Student Scholarship (each semester from Fall 2022 to Spring 2026, totally 6 times)
  Third Prize in the 7th National College Student Micro-Film Showcase – Long Video Category (03/2024)
  "Top 10 Psychological Counselors" of Chongqing University (07/2025)
- Student Work: School's Student Union (09/2022 – 09/2023), CQU-UC Mental Health Association (09/2022 – 09/2024)

University of Cincinnati Cincinnati, OH
B.S. (expected) in Electrical Engineering, CQU-UC Joint Co-op Institute *08/2026 – 06/2027*

RESEARCH EXPERIENCE

Research and Application of Hierarchical and Zonal Control Technology for Large-Scale Electric Vehicle Charging and Discharging – A National-level Smart Grid Research Project Chongqing, China*02/2025 – 06/2026*

- Focused on large-scale V2G coordination and microgrid planning. Developed an integrated optimization model that jointly addressed transformer thermal bottlenecks and battery cyclic aging, while utilizing a TSK fuzzy inference system to quantify user willingness to participate in V2G.
- Built MATLAB-based simulation and data-driven forecasting frameworks. Reformulated the complex MINLP model into a compact MIQP format via Big-M and convexification techniques to enable efficient large-scale co-optimization, solving seamlessly using YALMIP and Gurobi.
- Proposed a dynamic topology reconfiguration strategy by modeling MSB spatial topology as an active variable for V2G dispatch, overcoming conventional fixed-bus capacity limits. Validated with empirical data, significantly improving computational efficiency and cutting overall system operating costs by 62.6% under high V2G penetration.
- **PAPER**: **Zhou, M.**, Fang S., & Chen G. (2026). Optimal V2G Infrastructure Dispatch Considering Spatial MSB Configuration and Joint Asset Longevity. *IET Renewable Power Generation* (under review).

Multi-Objective Control Strategy and Evaluation System for Carbon Emission Mitigation Measures: Research and Patent Invention
Intern at the Institute of Hybrid Simulation, Tsinghua Sichuan Energy Internet Research Institute *03/2025 – 04/2025*

- Architected a five-module framework integrating surrogate modeling, HMOPSO optimization, and Kalman filter closed-loop evaluation. Drafted the technical disclosure for an institutional patent (featuring 3 core protection points), resolving the lack of dynamic carbon factor quantification and real-time adaptability in power dispatch.
- Utilized SCADA/EMS data and LSTM networks for accurate wind and solar load forecasting. Built a five-factor coordinated marginal carbon reduction surrogate model to optimize multi-source energy operations across thermal power, renewables, energy storage, and demand response.
- Designed a hierarchical multi-objective particle swarm optimization (HMOPSO) algorithm to efficiently identify the Pareto front for carbon emissions, costs, and reliability. Established a four-level dynamic closed-loop control system with rolling compensation, automatically triggering self-adjustments when execution efficiency falls below 80% to ensure operational reliability and lower carbon intensity.

AI-Based Power and Energy Balance Optimization System: Research and Patent Invention Chengdu, China
Intern at the Institute of Hybrid Simulation, Tsinghua Sichuan Energy Internet Research Institute *06/2024 – 07/2024*

- Led the design of a five-module optimization architecture to address real-time computational challenges in conventional power dispatch. Drafted the core technical disclosure for an institutional invention patent (featuring 2 key protection points), providing an integrated hardware-software solution for industrial-scale grid balancing.
- Integrated SARIMA and deep neural networks to extract time- and frequency-domain features of renewable energy fluctuations. Introduced an RBF-SVM model for high-dimensional anomaly detection and early warning hyperplane classification.

- Formulated a hybrid DQN-MILP optimization model to enable adaptive real-time dispatch. Deployed a distributed data acquisition and control system based on algorithmic simulations, significantly enhancing grid node response speed and overall dispatch accuracy.

Optimization of Servo Electric Cylinder Control System Based on LabVIEW and C++ Chongqing, China
Research Intern, State Key Laboratory of Mechanical Transmission for Advanced Equipment *01/2024 – 04/2024*

- Reconstructed the driver program on a DSP platform using C/C++ based on the FOC architecture, implementing Park/Clark transformations and SPWM/SVPWM modulation. Debugged the three-loop PID control (current, speed, and position) to significantly improve the dynamic response and control accuracy of the servo cylinder.
- Redesigned the LabVIEW-based human-machine interaction (HMI) system and integrated robust serial communication protocols for real-time parameter transmission and waveform acquisition, successfully resolving communication lag and data parsing packet loss issues during multi-parameter real-time monitoring.
- Integrated the DSP control algorithms with the LabVIEW HMI to establish a high-performance servo cylinder debugging platform, increasing the data refresh rate by nearly 30%. Wrote comprehensive technical documentation and user manuals to standardize the debugging process and ensure successful project acceptance.

INTERNSHIP EXPERIENCE

Chongqing Huayu Electrical Group Co., Ltd. Chongqing, China
R&D Testing and Technical Management Intern *at the Department of Inertial Navigation* *05/2026 – 07/2026*

- Developed MATLAB and Python scripts to automate CSV test-data processing, applied moving-average filtering to remove noise, and generated temperature-drift curves, supporting anomaly diagnosis and improving testing efficiency.
- Studied the operating principles of inertial sensors and translated 3 technical papers on MEMS inertial navigation packaging and emerging technologies, producing nearly 10,000 words of technical summaries to support R&D.
- Participated in the formatting and proofreading of technical reports under confidentiality regulations and GJB quality standards, checking figure numbering, formulas, and terminology for accuracy and consistency.
- Audited laboratory assets and checked precision instrument serial numbers and records; participated in technical review meetings and produced structured meeting minutes to facilitate cross-functional collaboration.

Chongqing University - University of Cincinnati Joint Co-op Institute Chongqing, China
Teaching Assistant *for Semiconductor Devices* and *Digital Design Courses* *09/2025 – 12/2025*

- Managed attendance, coordinated course enrollment, answered student inquiries, and organized laboratory sessions.
- Graded 17 theoretical assignments and 8 lab reports, providing feedback within 3 days, and reviewed 62 midterm exams and entered grades within 5 days.
- Compiled midterm course evaluation surveys and wrote a comparative analysis report (4~5 pages) of electrical engineering laboratory courses in China and the U.S., providing recommendations for course improvement.
- Supported the planning and on-site reception and execution of orientation activities, serving over 300 participants.

Sichuan Energy Internet Research Institute of Tsinghua University Chengdu, China
Power System Analysis Intern *at the Institute of Hybrid Simulation* *05/2024 – 08/2024 & 01/2025 – 04/2025*

- Code Refactoring and Simulation: Refactored and optimized C++ and MATLAB control code across languages and conducted renewable energy integration simulations and data analysis in Python. Optimized probabilistic wind and solar power output models based on MCMC, K-Means, and PSO, and developed an 8,760-hour load aggregation and dispatchable potential model for electrochemical energy storage in a regional power system.
- Academic Research and Papers: Drafted 4 chapters of the paper *Research on Protection Setting and Implementation of Back-to-Back VSC-HVDC Systems*. Gave an English presentation for the 4th International Conference on Power Systems and Energy Internet. Drafted the introduction of a research paper on renewable energy hosting capacity.
- Project Bidding and Technical Research: Wrote technical proposals for major projects including ultra-large urban power grids, flexible DC transmission, and AC-to-DC conversion of overhead transmission lines. Conducted research on power generation forecasting, cost and pricing, and grid dispatch mechanisms, and wrote technical reports.
- Technical Communication and Software Documentation: Delivered technical presentations on electromagnetic transient simulation platforms at power industry exhibitions in Chengdu and Guangzhou. Benchmarked BPA against dynamic electromagnetic transient simulation platforms using Simulink and PSCAD, and analyzed active/reactive power curves, applicability, and errors. Wrote a 26-page software user guide.

SKILLS & INTERESTS

- Language skills: Chinese (Native), English (Fluent), Korean (Basic)
- Computer skills: C++, MATLAB, Python, Jupyter, Simulink

- Interests: Foreign language, sports, reading, traveling